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tidyfinance: Transparent Factor Construction for Empirical Asset Pricing

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Abstract

The R package **tidyfinance** provides research infrastructure for the empirical asset pricing data workflow. This workflow turns raw accounting and return data into factor portfolios for asset pricing tests, optimal portfolio allocation, and the design of trading strategies. **tidyfinance** exposes every construction decision as a configurable parameter with a documented default. A unified `download_data()` interface covers WRDS (CRSP, Compustat), Ken French's Data Library, FRED, and other sources, and applies the cleaning, alignment, and delisting and survivorship corrections each provider requires. Modular functions then automate fiscal-year lagging, breakpoint computation, and portfolio aggregation. We illustrate the package by reproducing the asset growth factor end-to-end from raw data. Researchers who want precomputed series instead of rebuilding the pipeline can download a factor library of 50 sorting variables across 841,536 specifications with a single `download_data()` call or browse it at factors.tidy-finance.org.

Keywords: financial risk factors, reproducible research, asset pricing, R.

1. Introduction

Factor returns are essential inputs to empirical asset pricing tests, optimal portfolio allocation, and risk management decisions. Producing them is a three-stage pipeline: data acquisition (downloading and cleaning raw accounting and return data from CRSP, Compustat, and similar providers); variable construction (lagging variables to avoid look-ahead bias); and portfolio sorting (assigning securities to portfolios by characteristic). Each stage embeds construction choices: where to set breakpoints, whether to equal- or value-weight portfolios,

how to handle delistings, and which fiscal-year lag to apply. Few pipelines are standardised, and fewer still ship with executable code. Anyone who reproduces or extends a factor series therefore reconstructs the pipeline from scratch, with no way to verify a match to the original.

No widely adopted tool covers this upstream workflow. The most popular factor sources publish *series* without the code that produced them: Ken French's Data Library (Fama and French 2023) supplies the Fama-French three- and five-factor returns and dozens of sorted portfolios. Other projects do release code: the Open Source Bond Asset Pricing project (Dickerson, Mueller, and Robotti 2023, openbondassetpricing.com) covers fixed income, the Open Source Cross-Sectional Asset Pricing repository (Chen and Zimmermann 2022) reproduces 213+ published anomalies, and Jensen, Kelly, and Pedersen (2023) release a global factor library spanning hundreds of long-short signals across many countries. None of these sources, however, exposes the preprocessing pipeline as configurable code that researchers can inspect or modify. Many individual authors also host their own published factors, with transparency about construction choices varying widely. Existing R packages for quantitative finance work downstream of factor construction, on price and return panels the user must already have. Between raw data and standard asset pricing tests sits an opaque layer, where construction choices accumulate silently.

This paper introduces **tidyfinance**, an R package on CRAN that fills this gap. The package addresses each stage with a dedicated interface. A unified `download_data()` function cleans, aligns, and harmonises panels from WRDS (Wharton Research Data Services: CRSP, Compustat, TRACE, FISD), Ken French's Data Library, the Q5 factors of Hou, Mo, Xue, and Zhang (2021), the Open Source Asset Pricing repository, FRED, the Goyal-Welch macroeconomic predictors, Yahoo Finance, and current constituents for 20 major equity indexes. Composable helpers then automate fiscal-year lagging, breakpoint computation, and portfolio aggregation. Every construction choice appears as a function argument with a default that follows established convention (NYSE breakpoints, value-weights, July rebalancing, and the Fama-French fiscal-year lag). A default call reproduces standard practice; overriding any default takes one argument.

We illustrate the package by reproducing the asset growth factor of Cooper, Gulen, and Schill (2008) end-to-end from raw CRSP and Compustat data, then varying any construction choice with a single-argument change. To spare researchers the work of rebuilding the pipeline, we run **tidyfinance** across all defensible combinations and publish the results as a factor library of 50 sorting variables across 841,536 specifications. The library is available through a single `download_data()` call or an interactive dashboard at factors.tidy-finance.org. Users who prefer Python access the same interface through the companion package **tidyfinance** for Python. The package complements the open-access textbooks Scheuch, Voigt, and Weiss (2023) and Scheuch, Voigt, Weiss, and Frey (2024), which embed the workflow in pedagogical material for R and Python.

The remainder of the paper proceeds as follows. Section 2 places **tidyfinance** among existing libraries and packages. Section 3 walks through the package design and the three workflow stages. Section 4 illustrates the full pipeline on the asset growth factor. Section 5 documents the precomputed factor library and how to access it. Section 6 concludes.

2. Background and related work

The R ecosystem for empirical finance spans three layers, each with established package support but a missing piece between them.

Data acquisition. Several packages retrieve specific sources commonly used in empirical asset pricing. **frenchdata** (Areal 2021) downloads the Ken French data library. **quantmod** (Ryan and Ulrich 2025) and **tidyquant** (Dancho and Vaughan 2026) interface with Yahoo Finance, FRED, and other public price sources. **Rblpapi** (Armstrong, Eddelbuettel, Laing, and Ooms 2024) provides Bloomberg API access for subscribers. **highfrequency** (Boudt, Cornelissen, Payseur, Kleen, and Sjoerup 2024) handles intraday trade and quote data. None of these cover the WRDS-CRSP-Compustat panel that anchors most factor construction, nor do they implement the cleaning, alignment, and survivorship conventions the asset pricing literature requires.

Asset pricing tests and portfolio construction. Once a clean panel of returns is in hand, several packages support downstream analysis. **PerformanceAnalytics** (Peterson and Carl 2026) computes performance and risk metrics for return series. Portfolio optimisation is covered by **fPortfolio** (Wuertz, Setz, Chalabi, and Chen 2023). **RQuantLib** (Eddelbuettel, Nguyen, and Leitch 2024) interfaces with QuantLib for derivative pricing. Each of these tools assumes that the panel of returns and factors is already constructed.

Factor libraries. Recent work targets the upstream factor-construction step directly. Hou, Xue, and Zhang (2020) replicate 452 cross-sectional anomalies and document that most fail out-of-sample, attributing failures in part to undocumented construction variation across studies. Chen and Zimmermann (2022) respond by publishing code for 213+ anomaly portfolios, but their library targets a fixed, documented set rather than a reconfigurable pipeline. Jensen *et al.* (2023) provide a global factor database without an open construction workflow. None of the three exposes the preprocessing pipeline as configurable code that researchers can inspect or modify.

Tidy finance: transparency and modularity. tidyfinance sits between the data-acquisition layer and the downstream analysis packages, building on the workflow developed in *Tidy Finance with R* (Scheuch *et al.* 2023) and *Tidy Finance with Python* (Scheuch *et al.* 2024). The companion books embed the package in pedagogical material; this paper documents the package's design principles around factor construction, where the gap above is sharpest. Two pillars guide the package. *Transparency*: every construction decision is exposed as a configurable parameter with a documented default, following the *tidy data* discipline of Wickham (2014). *Modularity*: each construction step is a small composable function, so researchers can adopt the full pipeline by chaining functions or call individual pieces inside their own workflows. The current paper focuses on factor construction; the package itself covers further functionality for asset pricing tests and optimal portfolio choice that we defer to dedicated treatments.

3. Package overview

3.1. Installation and setup

You can install the released version of **tidyfinance** from CRAN via:

```
install.packages("tidyfinance")
```

To install the development version of **tidyfinance** from GitHub, first install the **pak** package if needed, then install the development version:

```
install.packages("pak")
pak::pak("tidy-finance/r-tidyfinance")
```

Once installed, load the package with:

```
library(tidyfinance)
```

3.2. Design and workflow

The empirical asset pricing workflow decomposes into three stages: data acquisition, variable construction, and portfolio sorting. Each stage involves dozens of small decisions: universe filters, cleaning rules, rebalancing conventions, breakpoint definitions. Figure 1 shows the pipeline and the core package functions at each stage. We discuss data acquisition, variable construction, and portfolio sorting in the following sections.

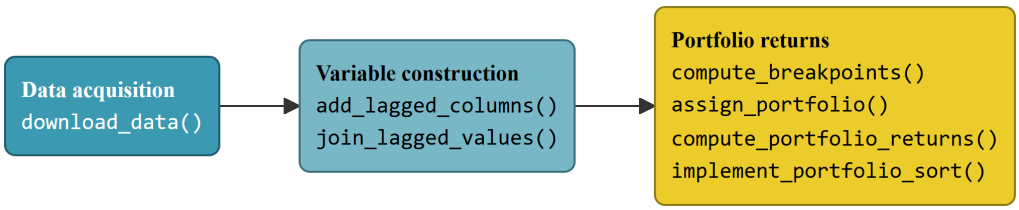


Figure 1: Research workflow for factor construction with **tidyfinance**, illustrating the three-stage pipeline and the core package functions involved at each step.

3.3. Data acquisition

tidyfinance provides a single interface, `download_data()`, to 321 datasets: equity returns, fixed-income transactions, macroeconomic predictors, index constituents, and more. Table 1 lists all available sources. `list_supported_datasets()` returns the full catalogue programmatically. The companion Python package **tidyfinance** exposes the same `download_data()` interface, with identical signatures and dataset identifiers, so analyses can be ported between the two languages without changes to the data layer.

Source	Description	Example
Public data sources		

Source	Description	Example

Ken French's Data Library	Fama-French 169+ factor models, portfolios, industry classifications	`download_data(`
		`  domain = "factors_ff",`
		`  dataset = "Fama/French 3 Factors"`
		`)`

Global Q	Hou-Xue-Zhang Q5 factor model (market, size, investment, ROE) at multiple frequencies	`download_data(`
		`  domain = "globalq",`
		`  dataset = "factors_q5_monthly"`
		`)`

Goyal-Welch Predictors	Macroeconomic return predictors	`download_data(`
		`  domain = "macro_predictors",`
		`  dataset = "monthly"`
		`)`

FRED	St. Louis Fed economic time series (GDP, CPI, etc.)	`download_data(`
		`  domain = "fred",`
		`  series = "GDP"`
		`)`

Yahoo Finance	Daily OHLCV with adjusted close for any ticker	`download_data(`
		`  domain = "stock_prices",`
		`  symbols = "AAPL"`
		`)`

Index Constituents	Current members of major equity indexes (S&P 500, Russell, DAX, MSCI World, etc.)	`download_data(`
		`  domain = "constituents",`
		`  index = "DAX"`
		`)`

Open Source Asset Pricing	213+ monthly asset pricing signals and anomalies	`download_data(`
		`  domain = "osap"`
		`)`

High-Frequency	5-second frequency data for S&P 500 constituents	`download_data(`
		`  domain = "tidyfinance",`
		`  dataset = "high_frequency_sp500"`
		`)`

Risk-Free Rate	Monthly risk-free rate (Fama-French)	`download_data(`
		`  domain = "tidyfinance",`
		`  dataset = "risk_free"`
		`)`

Factor Library	Precomputed factor returns across construction choices	`download_data(`
		`  domain = "tidyfinance",`
		`  dataset = "factor_library",`
		`  sorting_variable = "ag"`
		`)`

WRDS (subscription required)		

CRSP	Stock returns, market cap, exchange, industry	`download_data(`
		`  domain = "wrds",`
		`  dataset = "crsp_monthly"`
		`)`

Source	Description	Example

Compustat	Annual and quarterly financial statements	`download_data(`
		`  domain = "wrds",`
		`  dataset = "compustat_annual"`
		`)`

CCM Links	CRSP-Compustat linking table (permno to gvkey)	`download_data(`
		`  domain = "wrds",`
		`  dataset = "ccm_links"`
		`)`

FISD	Mergent Fixed Income Securities Database	`download_data(`
		`  domain = "wrds",`
		`  dataset = "fisd"`
		`)`

TRACE	Corporate bond transactions (Dick-Nielsen cleaned)	`download_data(`
		`  domain = "wrds",`
		`  dataset = "trace_enhanced",`
		`  cusips = "655844CJ5"`
		`)`

Table 1: Data sources available via `download_data()` in **tidyfinance**.

Most rows in Table 1 mirror upstream providers; the high-frequency dataset is the exception. We host it ourselves: it comprises message-level snapshots of the NASDAQ limit order book for the SPY exchange-traded fund (and TLT for government bonds) from July 2007 to October 2025, more than 30 billion order-book messages in total, curated for the analysis in Hautsch, Menkveld, and Voigt (2026).

A first illustration uses only public data sources. We retrieve the current constituents of the Dow Jones Industrial Average and then download daily prices for those tickers over a chosen window. `list_supported_indexes()` returns the full set of equity indexes available through the `"constituents"` domain.

```
dow_constituents <- download_data(
  domain = "constituents",
  index = "Dow Jones Industrial Average"
)

download_data(
  domain = "stock_prices",
  symbols = dow_constituents$symbol,
  start_date = "2020-01-01",
  end_date = "2024-12-31"
)
```

```
# A tibble: 37,710 x 8
   symbol date        volume  open   low  high close adjusted_close
```

```
   <chr>  <date>       <dbl> <dbl> <dbl> <dbl> <dbl>          <dbl>
 1 GS     2020-01-02 3736300  231   230.  235.  234.           202.
 2 GS     2020-01-03 2274500  232.  230.  233.  232.           200.
 3 GS     2020-01-06 3329300  230.  229.  234.  234.           202.
 4 GS     2020-01-07 5255200  235   235.  238.  235.           203.
 5 GS     2020-01-08 3564700  236.  235.  240.  238.           205.
 6 GS     2020-01-09 3980700  241.  240.  243.  243.           209.
 7 GS     2020-01-10 2248100  243.  241.  243.  242.           209.
 8 GS     2020-01-13 3359200  244.  243   246.  245.           212.
 9 GS     2020-01-14 4302800  245.  245.  249.  246.           212.
10 GS     2020-01-15 5411200  242   239.  250.  245.           212.
# i 37,700 more rows
```

Similarly, we can fetch the Fama-French three-factor data from Ken French's Data Library with a single call:

```
ff_factors <- download_data(
  domain = "factors_ff",
  dataset = "Fama/French 3 Factors"
)
```

```
No `start_date` or `end_date` provided. Returning the full data set.
```

The `start_date` and `end_date` arguments generalise across every dataset type in Table 1 that supports a time dimension.

Public data still require non-trivial cleaning. The Goyal-Welch (Goyal and Welch 2008) macroeconomic predictors are a typical case: `download_data(domain = "macro_predictors", ...)` fetches the raw spreadsheet at the requested frequency (monthly, quarterly, or annual), parses dates from the native `yyyymm`, `yyyyq`, or `yyyy` format, strips comma thousand separators, and coerces character columns to numeric. It then constructs the standard predictors used in the equity-premium prediction literature: the log dividend-price ratio (`dp`), dividend yield (`dy`), earnings-price ratio (`ep`), dividend-payout ratio (`de`), the forward equity risk premium (`rp_div`), the term spread (`tms`), and the default-yield spread (`dfy`). The output is the cleaned panel filtered to the requested date range.

For empirical asset pricing work, WRDS is the gold-standard source of return and accounting panels. WRDS data are proprietary, so an institutional subscription and credentials are required. **tidyfinance** handles this once: `set_wrds_credentials()` prompts for the username and password and stores them in a project-specific or global `.Renviron` file and additionally asks the user to add it to `.gitignore` to prevent credential leakage. Subsequent `download_data()` calls open the database connection automatically.

`download_data()` applies standard cleaning conventions following Scheuch *et al.* (2023) automatically. For CRSP monthly data, the function restricts the universe to common equity securities traded on NYSE, AMEX (now NYSE American), or NASDAQ. It computes market capitalisation as shares outstanding times price, constructs a one-month lag for portfolio weights, and maps exchange codes and SIC codes to readable labels. Excess returns are raw returns minus the Fama-French risk-free rate (the 1-month US T-bill); observations with missing returns or market capitalisation are dropped. Delisting returns are incorporated

directly, eliminating the need for a separate adjustment.

For Compustat annual data, `download_data()` restricts to industrial-format consolidated statements and retains only the most recent observation per firm-year. It constructs book equity in priority order: stockholders' equity when available, else common equity plus preferred stock, else total assets minus total liabilities. The function adds back deferred taxes and investment tax credits, and subtracts the book value of preferred stock.

For TRACE corporate bond transactions, `download_data()` applies the Dick-Nielsen cleaning protocol (Dick-Nielsen 2009) to remove cancellations, corrections, and reversals before returning the trade panel.

All datasets returned by `download_data()` use a single `date` column that follows the same rules across frequencies. Daily observations return the trading date. Monthly observations are dated to the last calendar day of the reference month, regardless of whether that day was a trading day. Quarterly and annual observations take the fiscal period-end date as reported by the original source; for Compustat, this is the firm's fiscal year-end (`datadate`), which is typically December of the fiscal year, but might differ across firms and within firms across years. Macroeconomic series from FRED keep their original release dates. A simple equality join on `date` therefore aligns observations of the same frequency, while cross-frequency joins, such as merging monthly returns with annual fundamentals, require the lagging logic described in the next subsection.

3.4. Variable construction

Most predictive variables in asset pricing studies are based on lagged information so that, for example, portfolio formation in month t always uses only data observable at the end of month $t-1$. `add_lagged_columns()` automates the required lagging procedure. For each row in the data, it looks back over a configurable window and attaches the most recent observed value of the chosen columns. Figure 2 illustrates the two modes the function supports. By default (`ff_adjustment = FALSE`), shown in Panel A, the window is bounded by `lag` (the minimum delay between observation and use) and `max_lag` (the maximum delay before a stale value is treated as missing). Within this window (the dashed rectangle), the function picks the most recent non-missing value; if no observation falls in the window, the lagged column takes `NA`.

The second mode, enabled by `ff_adjustment = TRUE` and shown in Panel B of Figure 2, implements the Fama-French convention (Fama and French 1993). Accounting variables for fiscal year $t-1$ are assumed to become public information by June of year t and are then used for portfolio formation from July of t through June of $t+1$. This produces step-function-lagged values that update once per year at the June/July boundary, regardless of the underlying fiscal year-end: all months from July through June share the same lagged value locked in at the previous rebalancing. In both modes, `add_lagged_columns()` selects the most recent observed value within the window.

`join_lagged_values()` is a related helper that merges lagged values from one data frame into another using the same windowing logic. It is useful when the target panel and the source panel are at different frequencies, for example when joining monthly accounting data into a quarterly target panel. The function takes the same `ff_adjustment` argument as

`add_lagged_columns()`, so the Fama-French convention can be applied at the join step rather than as a separate transformation.

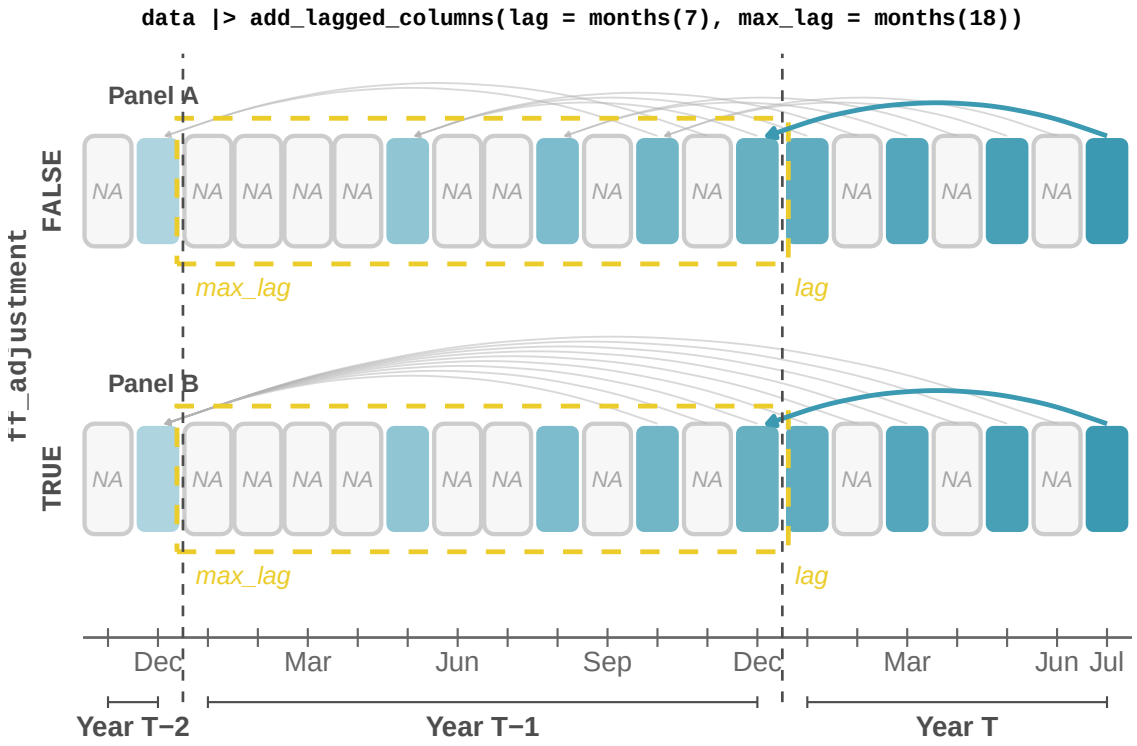


Figure 2: Two lagging mechanisms in `add_lagged_columns()`. Each row represents a data series with observed (filled) and missing (hollow) cells. The dashed rectangle marks the search window from the perspective of the most recent observation (the top row); arrows show which value gets matched to each target month. Panel A (`ff_adjustment = FALSE`) returns the most recent observed value within a window bounded by `lag` and `max_lag`. Panel B (`ff_adjustment = TRUE`) implements the Fama-French convention, where the lagged value updates only at the June/July boundary and is held constant for the next twelve months.

A handful of additional helpers cover routine operations that recur across empirical studies. `winsorize()` and `trim()` handle outliers in cross-sectional variables: the former symmetrically replaces values below the qth and above the $(1 - q)$th percentile with the corresponding quantile values, while the latter sets them to `NA`. `estimate_model()` fits a linear model from a formula string and returns the coefficients as a tidy one-row data frame, which composes naturally with `group_by()` for grouped or rolling regressions such as time-series factor exposures. `compute_rolling_value()` applies an arbitrary aggregation function over a rolling window defined in calendar time rather than row counts, accommodating irregular observation frequencies and missing periods.

3.5. Portfolio sorting

A portfolio sort groups stocks each period into ordered portfolios according to the level of a sorting variable. The boundaries between portfolios are called *breakpoints*: numeric cutoffs of the sorting variable that determine which portfolio a stock joins. A quintile sort on size, for instance, has four breakpoints at the 20th, 40th, 60th, and 80th percentiles of the cross-sectional market-capitalisation distribution. Common sorting variables include size, beta, book-to-market, and momentum. The long-short return between the extreme portfolios is interpreted as a factor-mimicking strategy and serves as the basis for tests of whether the sorting variable forecasts future returns. Portfolio sorts are among the most widely used tools in empirical asset pricing (Bali, Engle, and Murray 2016). We expose the procedure through four composable functions.

`compute_breakpoints()` takes a cross-section and a sorting variable and returns the numeric cutoffs that define the portfolio boundaries. The user supplies either a number of portfolios (for example, five for quintiles) or an explicit vector of percentiles, but not both. A `breakpoints_exchanges` argument restricts the universe used to compute breakpoints. The standard convention is NYSE-only, which prevents the large cross-section of small NASDAQ firms from dominating the lower bins. A related `breakpoints_min_size_threshold` argument excludes stocks below a market-cap quantile from the breakpoint computation, which is useful for size-related sorts where microcaps would otherwise distort the cutoffs. A `smooth_bunching` flag handles cases where the sorting variable clusters at the edges by pinning extreme values to the boundary portfolios and recomputing breakpoints on the remainder.

`assign_portfolio()` maps each observation to a portfolio integer and uses breakpoints supplied either directly or via a breakpoint-computing function (defaulting to `compute_breakpoints()`). In practice, `assign_portfolio()` is called separately for each rebalancing date, so that portfolios are reformed each period rather than fixed once for the whole sample.

`compute_portfolio_returns()` ties the lower-level pieces together into a single call. Given a panel of stock data, one or two sorting variables, a sorting method, and `breakpoint_options`, it handles rebalancing, breakpoint computation, portfolio assignment, and return aggregation. It then returns a complete panel of portfolio returns. For each portfolio-date the function reports three return measures: equal-weighted, value-weighted using `mktcap_lag` as the weight, and capped value-weighted (Jensen *et al.* 2023). The capped version pulls weights above the `cap_weight` percentile (default 0.8) back to that cap, which prevents a small number of megacap firms from dominating the portfolio. A `min_portfolio_size` argument sets the minimum number of constituents below which the corresponding return is set to `NA`. An optional `rebalancing_month` argument switches from period-by-period rebalancing to annual rebalancing in the specified month. Portfolios formed in that month are then held constant for the following twelve months.

The function supports three sorting schemes, illustrated in Figure 3. A *univariate* sort uses a single variable. An *independent bivariate* sort computes breakpoints on each variable separately and intersects the resulting bins to produce a two-way grid. A *dependent bivariate* sort first partitions the cross-section by the secondary (control) variable and then re-sorts within each bucket on the main variable, which is the appropriate choice when the second variable is treated as a control rather than as an independent dimension of variation. For bivariate sorts, returns are averaged over the secondary variable so that only the main-variable portfolios are returned.

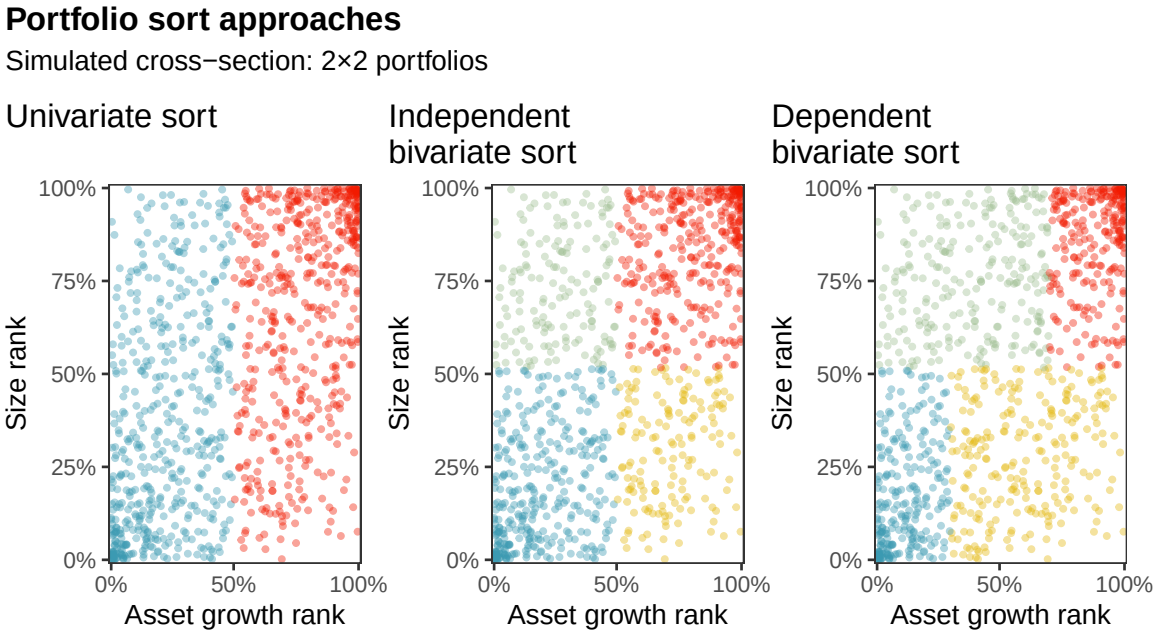


Figure 3: Portfolio sort approaches illustrated on a simulated cross-section. Each panel plots simulated observations assigned to 2×2 portfolios based on asset growth rank (x-axis) and size rank (y-axis), both expressed as percentiles. Colours indicate portfolio cell assignment. The univariate sort (left) assigns portfolios on asset growth alone; the independent bivariate sort (centre) assigns each dimension independently; the dependent bivariate sort (right) sorts on size first, then on asset growth within size groups.

`compute_long_short_returns()` is a convenience wrapper that takes the output of `compute_portfolio_returns()` and computes the spread between the top and bottom portfolios. A `direction` argument controls the sign and defaults to `"top_minus_bottom"`. The function returns one long-short series per return measure available in the input, so the equal-weighted, value-weighted, and capped value-weighted spreads are produced in a single call. This long-short return is the factor-mimicking spread that the asset-pricing literature typically takes as the object of interest.

For users who want the full pipeline in a single call, `implement_portfolio_sort()` wraps the four functions above. It takes a panel of stock data, one or two sorting variables, a sorting method, and a configuration object, and returns the long-short factor return series alongside the underlying portfolio returns. The lower-level functions remain available for custom workflows that need finer control over breakpoints, weighting, or rebalancing.

4. A case study in factor construction

We walk through the construction of the asset growth factor from raw CRSP and Compustat data to long-short portfolio returns, applying each stage of Figure 1 in turn. Each step uses a **tidyfinance** function and generalises directly to any sorting variable in the factor library.[1]

Once installed, load **tidyfinance** together with a minimal set of supporting packages: **dplyr** (Wickham, François, Henry, Müller, and Vaughan 2026) for data manipulation, **tidyr** (Wickham, Vaughan, and Girlich 2024) for reshaping between long and wide formats, and **lubridate** (Grolemund and Wickham 2011) for date arithmetic.

```
library(tidyfinance)
library(dplyr)
library(tidyr)
library(lubridate)
```

4.1. Data download

Constructing the asset growth factor requires two WRDS panels: annual accounting data from Compustat to compute the asset growth signal and monthly stock returns from CRSP to translate the signal into portfolio returns. Before the first WRDS download, call `set_wrds_credentials()` to store WRDS credentials securely in `.Renviron`; subsequent `download_data()` calls establish the database connection automatically.

```
date_start <- as.Date("1960-01-01")
date_end <- as.Date("2024-12-31")
```

First, we download monthly stock returns from CRSP. Setting `add_ccm_links = TRUE` appends CRSP-Compustat linking identifiers needed to merge with financial statement data in the next step.

```
crsp_monthly <- download_data(
  domain = "wrds",
```

[1] Installation of **tidyfinance** is described in Section 3.1.

```
  dataset = "crsp_monthly",
  version = "v2",
  start_date = date_start,
  end_date = date_end,
  add_ccm_links = TRUE
)
```

The dataset contains 3,451,594 stock-month observations for 26,263 unique stocks, covering the period from January 1st, 1960 to December 1st, 2024.

Next, we download annual financial statements from Compustat.

```
compustat_annual <- download_data(
  domain = "wrds",
  dataset = "compustat_annual",
  start_date = date_start,
  end_date = date_end,
  additional_columns = c("at", "ib"),
  only_usd = TRUE
) |>
  select(gvkey, date, at, ib)
```

Annual Compustat data contains 506,808 firm-year observations for 39,633 unique firms, covering the period from January 1st, 1960 to December 31st, 2024. The `additional_columns` argument retrieves total assets (`at`) and income before extraordinary items (`ib`), needed for computing asset growth and applying an earnings filter in later steps. Setting `only_usd = TRUE` restricts the dataset to USD-denominated stocks, ensuring consistency with CRSP.

4.2. Data cleaning and variable construction

We illustrate data cleaning and variable construction using asset growth (AG) as our running example. Asset growth is the year-over-year percentage change in total assets (Cooper *et al.* 2008),

$$AG_t = (AT_t - AT_{t-1})/AT_{t-1},$$

where AT_t is total assets in year t (Compustat field `at`).

The `add_lagged_columns()` function from **tidyfinance** handles temporal alignment by creating a lagged version of total assets grouped by firm identifier (`gvkey`) with a one-year lag. Non-finite values (arising when lagged total assets are zero or missing) become `NA`.

```
sorting_variable_data <- compustat_annual |>
  add_lagged_columns(
    cols = "at",
    lag = years(1),
    by = "gvkey",
  ) |>
  mutate(
```

```
    asset_growth = (at - at_lag) / at_lag,
    asset_growth = if_else(is.finite(asset_growth), asset_growth, NA_real_)
  ) |>
  select(gvkey, date, asset_growth, ib)
```

The call relies on two silent defaults: `max_lag` falls back to `lag` (here one year), so only the single observation exactly one year before each row is eligible; `ff_adjustment = FALSE`, so the look-back is the plain calendar window rather than the Fama-French June/July rebalancing convention introduced in Section 3.4.

`sorting_variable_data` contains one observation per firm-year, the natural unit for cross-sectional analysis of annual accounting variables. Asset growth is heavily right-skewed: a small number of firms with dramatic balance-sheet expansions generate extreme positive values. Table 2 reports the pooled cross-sectional summary alongside the time-series average, minimum, and maximum of the corresponding statistic computed cross-sectionally each year.

```
pooled_stats <- sorting_variable_data |>
  drop_na(asset_growth) |>
  mutate(asset_growth = winsorize(asset_growth, 0.01)) |>
  summarise(
    N = n(),
    Mean = mean(asset_growth),
    SD = sd(asset_growth),
    P10 = quantile(asset_growth, 0.10),
    P25 = quantile(asset_growth, 0.25),
    Median = median(asset_growth),
    P75 = quantile(asset_growth, 0.75),
    P90 = quantile(asset_growth, 0.90)
  )

yearly_stats <- sorting_variable_data |>
  drop_na(asset_growth) |>
  group_by(year = year(date)) |>
  mutate(
    p01 = quantile(asset_growth, 0.01),
    p99 = quantile(asset_growth, 0.99),
    asset_growth = pmax(pmin(asset_growth, p99), p01)
  ) |>
  summarise(
    N = n(),
    Mean = mean(asset_growth),
    SD = sd(asset_growth),
    P10 = quantile(asset_growth, 0.10),
    P25 = quantile(asset_growth, 0.25),
    Median = median(asset_growth),
    P75 = quantile(asset_growth, 0.75),
    P90 = quantile(asset_growth, 0.90),
    .groups = "drop"
```

Statistic	Pooled	Yearly avg	Yearly min	Yearly max
N	389,127	6,080	1,488	9,467
Mean	0.238	0.242	0.048	0.889
SD	0.858	0.847	0.173	3.830
P10	-0.191	-0.164	-0.397	0.007
P25	-0.034	-0.027	-0.151	0.060
Median	0.064	0.067	0.001	0.132
P75	0.207	0.207	0.105	0.370
P90	0.588	0.590	0.253	1.341

Table 2: Cross-sectional summary statistics for asset growth (winsorised at the 1st and 99th percentiles). The 'Pooled' column reports the statistic computed on the entire firm-year sample. The 'Yearly' columns summarise the time series of statistic values computed cross-sectionally year by year, taking the average, minimum, and maximum across calendar years. N is the count of firm-year observations (pooled) or per-year observations (yearly columns).

```
  ) |>
  select(-year)
```

Table 2 assembles the two computations side by side.

The next step merges annual sorting variables onto the monthly CRSP stock panel using `join_lagged_values()`. The function aligns fiscal-year-end data with monthly observations through a configurable lag window. Setting `ff_adjustment = TRUE` implements the Fama-French convention (Fama and French 1993), aligning fiscal-year data from year t with portfolio formation in July of year $t+1$ through a 12-to-18-month look-back window. The same lagging process generalises to all sorting variables in the full factor library, and the factor library API provides precomputed lagged panels for all of them. Specifically, we implement three- and six-month lags with maximum look-back periods of 15 and 18 months, respectively.

Figure 4 shows the 33rd, 50th, and 66th percentiles of asset growth over time in the resulting panel; pronounced spikes during economic expansions and contractions highlight the cyclical nature of corporate investment that portfolio sorts aim to exploit.

```
sorting_data <- crsp_monthly |>
  join_lagged_values(
    new_data = sorting_variable_data,
    id_keys = "gvkey",
    min_lag = months(7),
    max_lag = months(18),
    ff_adjustment = TRUE
  )

sorting_data <- sorting_data |>
  select(
    date,
    permno,
```

```
    exchange,
    siccd,
    ret_excess,
    mktcap_lag,
    asset_growth,
    ib
  ) |>
  filter(date > date_start + months(12) + months(18))
```

The filter removes the initial year used to compute the growth rate plus an additional 18 months to avoid mechanically introduced gaps from the lagging step, leaving a panel of stock-month observations with excess returns, exchange membership, lagged market capitalisation, the lagged sorting variable. The auxiliary columns `siccd` and `ib` support the sample-construction filters in Section 4.3.2.

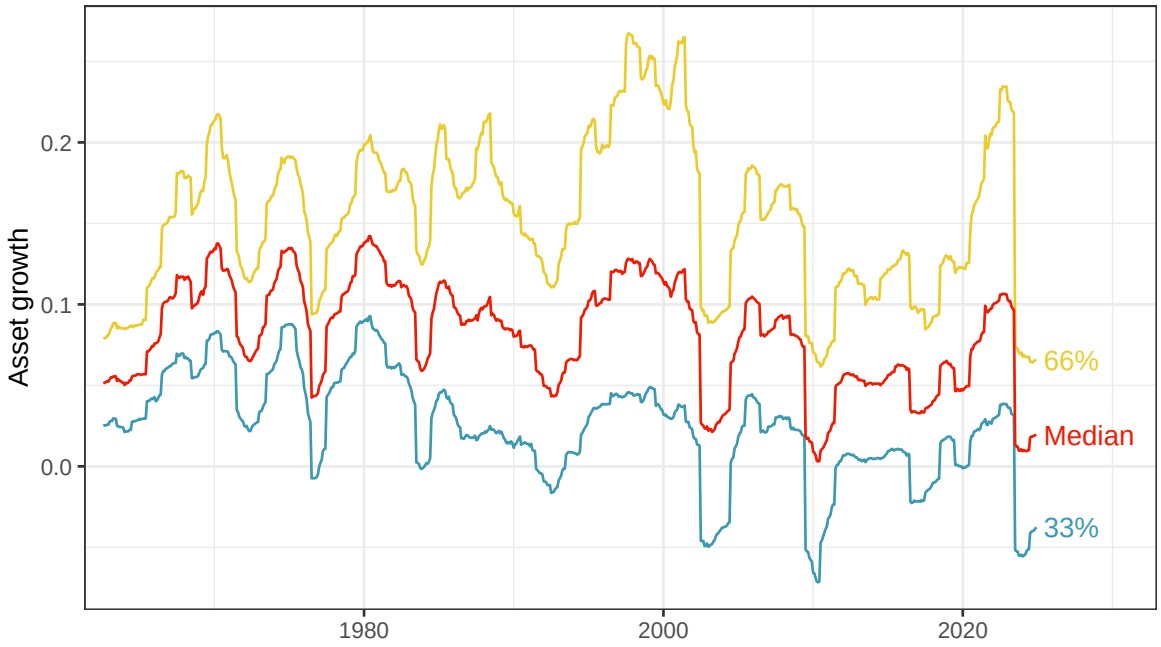


Figure 4: Selected percentiles (33rd, 50th, 66th) of cross-sectional asset growth over time. Asset growth is expressed as a proportion (e.g., 0.2 = 20% annual growth in total assets) and covers the full sample period of the merged CRSP-Compustat dataset.

With the sorting variable constructed and merged onto the monthly stock panel, we now turn to portfolio sorts, the mechanism that translates firm-level characteristics into tradable factor returns.

4.3. Portfolio sorts

In portfolio sorts, stocks are ranked each period by their sorting variable and divided into quantile portfolios. Portfolio returns are equal-weighted or value-weighted averages of constituent stock returns. The long-short factor return is the difference between the extreme portfolios.

For asset growth, the relevant spread is bottom minus top, reflecting the negative premium documented by Cooper *et al.* (2008). The construction involves several methodological choices: which stocks enter the universe, how many portfolios to form, which exchange determines the breakpoints, the weighting scheme, and the rebalancing frequency.

`implement_portfolio_sort()` exposes all of these choices through a single function call. The function takes the panel data, one or two sorting variables, a sorting method, and a `portfolio_sort_options` object that bundles two configuration layers: `filter_options()` controls which observations enter the breakpoint and return computation (for example, excluding financial firms or stocks below a minimum price), and `breakpoint_options()` controls how portfolios are formed (number of portfolios, exchange universe, bunching). Internally, the function applies the sample filters and then calls `compute_portfolio_returns()`, which produces the time series of portfolio returns. A subsequent call to `compute_long_short_returns()` constructs the hedge portfolio.

`compute_portfolio_returns()` itself rests on two lower-level helpers, `compute_breakpoints()` for quantile cut-points and `assign_portfolio()` for mapping each stock-month to a portfolio number. Both are exported, so researchers who need finer control can call them directly rather than through the wrapper. The two examples below use `implement_portfolio_sort()` throughout.

Baseline specification (univariate sorts)

The simplest call to `implement_portfolio_sort()` requires only a `portfolio_sort_options` object that specifies how to form the breakpoints. In a univariate sort, stocks are ranked on a single characteristic. We illustrate this here using asset growth as the sorting variable and forming quintile portfolios with breakpoints computed from NYSE-listed stocks only, and rebalanced annually in July.

```
options_baseline <- portfolio_sort_options(
  breakpoint_options_main = breakpoint_options(
    n_portfolios = 5,
    breakpoints_exchanges = "NYSE"
  )
)
```

The `breakpoint_options()` call specifies that breakpoints come from NYSE-listed stocks only, a common convention that avoids the influence of small AMEX and NASDAQ firms on portfolio cutoffs. We then pass these options to `implement_portfolio_sort()`.

```
portfolio_returns <- implement_portfolio_sort(
  data = sorting_data,
  sorting_variables = "asset_growth",
  sorting_method = "univariate",
  rebalancing_month = 7,
  portfolio_sort_options = options_baseline
)
```

Apart from setting `rebalancing_month = 7`, every other choice point falls back to the package

default: no smooth bunching, no minimum portfolio size, and a cap-weight cutoff at the 80th percentile of market capitalisation for the capped value-weighted return. The standard CRSP/Compustat column names (`permno`, `date`, `ret_excess`, `mktcap_lag`, `exchange`) are picked up automatically through the default `data_options()`. Setting `rebalancing_month = 7` reforms the portfolios each July and holds them for the following twelve months, the standard Fama-French rebalancing convention. This pairs with `ff_adjustment = TRUE` in `join_lagged_values()`: the signal is locked in at the June/July boundary, so July is the first month in which a fresh fiscal-year update is available.

`compute_portfolio_returns()`, called internally, returns a long tibble with one row per portfolio-month and the three weighting columns introduced in Section 3.5 (`ret_excess_vw` for value-weighted returns (VW), `ret_excess_ew` for equal-weighted returns (EW), and `ret_excess_vw_capped` for capped value-weighted returns (VWC)). Piping the result through `compute_long_short_returns()` gives the hedge portfolio. The `direction = "bottom_minus_top"` argument means the factor is long low-AG stocks and short high-AG stocks, in line with the negative premium documented by Cooper *et al.* (2008).

```
factor_returns_ag <- portfolio_returns |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

Table 3 summarises the time series of each quintile across all three weighting schemes, with the long-short portfolio appended as the final row.

```
quintile_stats <- portfolio_returns |>
  drop_na() |>
  group_by(portfolio) |>
  summarise(
    vw_mean = mean(ret_excess_vw) * 12,
    vw_tstat = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n())),
    ew_mean = mean(ret_excess_ew) * 12,
    ew_tstat = mean(ret_excess_ew) / (sd(ret_excess_ew) / sqrt(n())),
    vwc_mean = mean(ret_excess_vw_capped) * 12,
    vwc_tstat = mean(ret_excess_vw_capped) /
      (sd(ret_excess_vw_capped) / sqrt(n())),
    .groups = "drop"
  ) |>
  mutate(portfolio = as.character(portfolio))

ls_row <- factor_returns_ag |>
  drop_na() |>
  summarise(
    portfolio = "L/S",
    vw_mean = mean(ret_excess_vw) * 12,
    vw_tstat = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n())),
    ew_mean = mean(ret_excess_ew) * 12,
    ew_tstat = mean(ret_excess_ew) / (sd(ret_excess_ew) / sqrt(n())),
    vwc_mean = mean(ret_excess_vw_capped) * 12,
    vwc_tstat = mean(ret_excess_vw_capped) /
      (sd(ret_excess_vw_capped) / sqrt(n()))
```

Portfolio	VW Mean	VW t	EW Mean	EW t	VWC Mean	VWC t
1	0.091	4.281	0.147	4.770	0.102	4.028
2	0.078	4.265	0.116	5.077	0.094	4.461
3	0.069	3.718	0.107	4.781	0.090	4.355
4	0.072	3.605	0.101	4.238	0.086	3.870
5	0.066	2.610	0.048	1.583	0.054	1.930
L/S	0.026	1.924	0.099	9.116	0.047	4.468

Table 3: Time-series averages of annualised monthly excess returns for asset growth quintile portfolios. Portfolios are formed annually in July using NYSE breakpoints. Means are annualised (multiplied by 12). The t-statistic tests whether the mean monthly excess return differs from zero. VW denotes value-weighted, EW equal-weighted, and VWC capped value-weighted (weights above the 80th percentile pulled back to that cap). The L/S row is the long-short portfolio (bottom minus top quintile).

```
)

univariate_sort_table <- bind_rows(quintile_stats, ls_row)
```

The value-weighted long-short portfolio earns 2.6% per year (t-statistic of 1.92), consistent with the negative premium documented by Cooper *et al.* (2008). The equal-weighted spread is larger, as expected for characteristics that correlate with size. The monotone decline of value-weighted means across quintiles confirms that `tidyfinance` reproduces the literature result from raw WRDS data with a single function call.

Customised specification (bivariate sorts)

The baseline specification above takes (almost) every package default and runs a univariate sort to form long-short portfolios. The customised specification below changes both: it overrides several of those defaults through `portfolio_sort_options` and moves from sorting on one variable to sorting on two.

We adopt a specification that combines, in a single call to `implement_portfolio_sort()`, the universe restrictions and breakpoint conventions used in many empirical studies: exclude financials and utilities, exclude firms with non-positive earnings, ignore the smallest 20% of NYSE-sized firms when computing breakpoints, form quintiles on AG within size buckets, use NYSE-only breakpoints, rebalance annually in July, and require at least ten constituents per portfolio.

Bivariate (double) sorts control for a second variable, typically firm size (`mktcap_lag`), when forming portfolios on the main characteristic. We use the dependent variant introduced in Section 3.5 and illustrated in Figure 3: stocks are first split into size buckets, then sorted into asset growth portfolios within each bucket, with returns averaged across the size dimension.

```
options_customised <- portfolio_sort_options(
  filter_options = filter_options(
    exclude_financials = TRUE,
```

```
    exclude_utilities = TRUE,
    exclude_negative_earnings = TRUE
  ),
  breakpoint_options_main = breakpoint_options(
    n_portfolios = 5,
    breakpoints_exchanges = "NYSE",
    breakpoints_min_size_threshold = 0.2
  ),
  breakpoint_options_secondary = breakpoint_options(
    n_portfolios = 2,
    breakpoints_exchanges = "NYSE"
  )
)

portfolio_returns_customised <- implement_portfolio_sort(
  data = sorting_data,
  sorting_variables = c("asset_growth", "mktcap_lag"),
  sorting_method = "bivariate-dependent",
  rebalancing_month = 7,
  portfolio_sort_options = options_customised,
  min_portfolio_size = 10,
  quiet = TRUE
)

factor_returns_customised <- portfolio_returns_customised |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

A single object-construction call (`options_customised`) bundles every methodological choice that distinguishes this specification from the baseline; nothing in the surrounding pipeline changes. Switching from a univariate to a size-controlled dependent double sort is a one-word change in `sorting_method`, and the secondary breakpoint object is supplied alongside the primary one without any additional work. Setting `quiet = TRUE` suppresses the diagnostic messages that `filter_sorting_data()` emits reporting how many observations each filter drops; setting `quiet = FALSE` (the default) is useful when iterating to track sample-construction effects.

For comparison, we also include a bivariate-dependent sort that uses the default filters and breakpoint conventions, so the only difference from the baseline is the addition of size control. The long-short returns from that specification are appended to the customised one in Table 4.

```
options_bivariate <- portfolio_sort_options(
  breakpoint_options_main = breakpoint_options(
    n_portfolios = 5,
    breakpoints_exchanges = "NYSE"
  ),
  breakpoint_options_secondary = breakpoint_options(
    n_portfolios = 2,
    breakpoints_exchanges = "NYSE"
  )
```

```
)

portfolio_returns_bivariate <- implement_portfolio_sort(
  data = sorting_data,
  sorting_variables = c("asset_growth", "mktcap_lag"),
  sorting_method = "bivariate-dependent",
  rebalancing_month = 7,
  portfolio_sort_options = options_bivariate
)

factor_returns_bivariate <- portfolio_returns_bivariate |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

Table 4 decomposes the customised premium into two layers. Adding the size-controlled double sort alone (the bivariate-only row) lifts the annualised premium from 2.6% to 3.6%, with the t-statistic moving from borderline to clearly significant. Controlling for size therefore reveals that most of the AG signal in the cross-section was masked by size dispersion in the univariate sort. Layering the filter set and the 20% breakpoint threshold on top trims the magnitude back to 3%, where the spread remains significant at conventional levels: removing financials, utilities, and loss-making firms strips out some of the firms that were contributing to the raw spread. Across all three specifications, the spread is positive and reflects the negative AG premium of Cooper *et al.* (2008).

```
customised_comparison_table <- bind_rows(
  factor_returns_ag |>
    drop_na() |>
    summarise(
      Specification = "Baseline (univariate, no filters)",
      `VW Mean` = mean(ret_excess_vw) * 12,
      `VW $t$` = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n()))
    ),
  factor_returns_bivariate |>
    drop_na() |>
    summarise(
      Specification = "Bivariate only (size-controlled, no filters)",
      `VW Mean` = mean(ret_excess_vw) * 12,
      `VW $t$` = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n()))
    ),
  factor_returns_customised |>
    drop_na() |>
    summarise(
      Specification = "Customised (size-controlled, filtered)",
      `VW Mean` = mean(ret_excess_vw) * 12,
      `VW $t$` = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n()))
    )
)
```

For researchers who want even finer control, the package exposes the lower-level build-

Specification	VW Mean	VW t
Baseline (univariate, no filters)	0.026	1.924
Bivariate only (size-controlled, no filters)	0.036	3.376
Customised (size-controlled, filtered)	0.030	2.439

Table 4: Comparison of long-short asset growth returns across three specifications. The baseline is a univariate sort with NYSE breakpoints and no filters. The bivariate-only specification adds a dependent double sort on size to the baseline but otherwise keeps the same universe and breakpoints. The customised specification adds the filter set (financials, utilities, non-positive earnings) and a 20% size threshold on the breakpoints. All rows use annual July rebalancing; means are annualised value-weighted excess returns.

ing blocks (`filter_sorting_data()`, `compute_breakpoints()`, `assign_portfolio()`, `compute_portfolio_returns()`) as standalone functions. They share signatures and semantics with `implement_portfolio_sort()`, so an analysis that prototypes against the wrapper can be unrolled into its components without rewriting the pipeline. This separation is convenient when custom logic, for example, computing breakpoints separately within each industry, or from a trailing window of past data rather than the contemporaneous cross-section, needs to be applied at one of the stages only.

Asset pricing tests

A long-short return is one summary of whether asset growth predicts the cross-section of returns. A complementary view comes from Fama-MacBeth regressions (Fama and MacBeth 1973), which estimate the average risk premium attached to a characteristic after running cross-sectional regressions period by period. `tidyfinance` implements the standard procedure in `estimate_fama_macbeth()`, which takes a panel and a model formula, fits one cross-sectional OLS regression per period, and aggregates the period-level coefficients into average risk premia with Newey-West (Newey and West 1987) standard errors with automatic lag selection.

Here, we winsorise the AG signal at the 1st and 99th percentiles to keep extreme observations from dominating the cross-sectional regressions and pair it with log market capitalisation as a size control.

```
fm_data <- sorting_data |>
  drop_na(asset_growth, mktcap_lag, ret_excess) |>
  group_by(date) |>
  mutate(
    asset_growth = winsorize(x = asset_growth, cut = 0.01),
    log_mktcap = log(mktcap_lag)
  ) |>
  ungroup()

fm_results <- estimate_fama_macbeth(
  data = fm_data,
  model = "ret_excess ~ asset_growth + log_mktcap"
)
```

Variable	Risk premium	SE	t-stat	N (months)
Intercept	0.0158	0.0042	3.7764	749
Asset growth	-0.0065	0.0009	-7.3861	749
Log market cap	-0.0015	0.0004	-3.4271	749

Table 5: Fama-MacBeth regression of monthly excess returns on lagged asset growth and lagged log market capitalisation. Coefficients are time-series averages of monthly cross-sectional OLS estimates; standard errors are Newey-West (Newey and West 1987) with automatic lag selection. The intercept and slopes on the right-hand-side variables are reported on the monthly scale.

The negative slope on `asset_growth` lines up with the long-short evidence in Table 3 and Table 4: high-AG firms earn lower subsequent returns, conditional on size. Switching to `vcov = "iid"` or supplying `vcov_options = list(lag = 6, prewhite = FALSE)` would change the standard-error treatment without altering the rest of the call, and adding additional regressors (a beta estimate from `estimate_betas()`, a profitability measure, an idiosyncratic-volatility control) is a one-line extension to the model formula. The same function therefore covers the standard cross-sectional asset-pricing test that complements the portfolio-sort evidence used throughout the rest of the paper.

5. A transparent risk factor library

5.1. Specification grid

When every construction choice is configurable, no single combination is canonical. Rather than ship one default series and force users to rebuild the pipeline for every variant, we run **tidyfinance** across all defensible combinations and publish the results as a precomputed factor library. Walter, Weber, and Weiss (2024) and Gruber, Kastner, Voigt, and Weiss (2025), among others, argue for systematic enumeration over construction choices in preference to any single canonical specification.

The library is constructed by running each combination of the choices in Table 6 on the same underlying CRSP-Compustat universe. Beyond the variation Table 6 makes explicit, two universe filters apply across all specifications. An *earnings filter* drops firms with non-positive income before extraordinary items (`ib`), which is standard in the literature to mitigate the influence of financial distress on the resulting portfolios. An *age filter* drops firms younger than two years, which ensures the sorting variable is based on stable financial information. Both filters are exposed as parameters of `download_data()`; users who want to switch them off can override the default.

Table 6 lists the preprocessing choices that vary across the library. Italic values mark the default for each choice; non-italic values are the alternatives we run.

Preprocessing choice	Argument	Options	Description
Size exclusion	`min_size_quantile`	None; *0.2*	Fraction of the smallest stocks (by market cap) excluded from the portfolio universe.
Exclusion of financials	`exclude_financials`	*No*; Yes	Drops financial-sector stocks (SIC 6000–6999) from the universe.
Exclusion of utilities	`exclude_utilities`	*No*; Yes	Drops utility-sector stocks (SIC 4900–4999) from the universe.
Exclusion of neg. earnings	`exclude_negative_earnings`	*No*; Yes	Drops firms with negative earnings before sorting.
Sorting variable lag	`sorting_variable_lag`	3m; *6m*; FF	Lag applied to the sorting variable before portfolio assignment (e.g., 6m means a 6-month lag).
Rebalancing	`rebalancing`	*Monthly*; Annual	How frequently portfolios are reformed.
Quantile groups (main)	`n_portfolios_main`	3; 5; *10*	Number of quantile groups (e.g., 10 for decile portfolios).
Sort method	`sorting_method`	*Univariate*; Bivariate (dep.); Bivariate (indep.)	Single sort, or dependent or independent double sort on size.
Quantile groups (secondary)	`n_portfolios_secondary`	*NA*; 2; 5	Number of groups for the secondary (size) sort dimension; NA for univariate sorts, 2 or 5 for bivariate sorts.
Breakpoint exchanges	`breakpoints_exchanges`	*NYSE*; All	Exchange(s) used to compute breakpoints. NYSE-only follows the conventional Fama-French approach.

Breakpoint min size threshold	`breakpoints_min_size_threshold`	*NA*; 0.2	Minimum market-cap quantile applied when computing breakpoints; NA disables the screen.
Weighting scheme	`weighting_scheme`	EW; *VW*	Return weighting within portfolios: equal-weighted (EW) or value-weighted (VW).

Table 6: Preprocessing choices in the portfolio sort grid and corresponding `download_data()` arguments. Italic values indicate the default. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Fama and French (1993).

The factor library covers 50 sorting variables across seven categories: investment, profitability, valuation, momentum, intangibles, trading frictions, and financing. The selection follows Jensen *et al.* (2023). Table 7 lists all variables with their long-short direction and primary reference. Each variable runs through every valid combination of the choices in Table 6, producing 841,536 factor return series in total.

Variable	Name		Main reference
52w	52-Week High	↑	George and Hwang (2004)
aci	Abnormal Corporate Investment	↓	Titman, Wei, and Xie (2004)
ami	Amihud Illiquidity	↑	Amihud (2002)
ag	Asset Growth	↓	Cooper *et al.* (2008)
beta	Beta	↑	Fama and MacBeth (1973)
bm	Book-to-Market	↑	Daniel and Titman (2006)
cfm	Cash Flow to Market	↑	Lakonishok, Shleifer, and Vishny (1994)
cbop	Cash-Based Operating Profitability	↑	Ball, Gerakos, Linnainmaa, and Nikolaev (2016)
dbe	Change in Common Equity	↓	Richardson, Sloan, Soliman, and Tuna (2005)
dcol	Change in Current Operating Liabilities	↓	Richardson *et al.* (2005)
dfnl	Change in Financial Liabilities	↓	Richardson *et al.* (2005)
dnoa	Change in Net Operating Assets	↓	Hirshleifer, Hou, Teoh, and Zhang (2004)
dpia	Change in PPE and Intangibles	↓	Lyandres, Sun, and Zhang (2008)
dwc	Change in Working Capital	↓	Richardson *et al.* (2005)

cdi	Composite Debt Issuance	↓	Lyandres *et al.* (2008)
csi	Composite Share Issuance	↓	Daniel and Titman (2006)
dm	Debt to Market	↑	Bhandari (1988)
dtv	Dollar Trading Volume	↓	Brennan, Chordia, and Subrahmanyam (1998)
em	Earnings to Market	↑	Basu (1983)
ebm	Enterprise Book to Market	↑	Penman, Richardson, and Tuna (2007)
gpa	Gross Profits to Assets	↑	Novy-Marx (2013)
hr	Hiring Rate	↓	Belo, Lin, and Bazdresch (2014)
iskew	Idiosyncratic Skewness	↓	Bali *et al.* (2016)
ivol	Idiosyncratic Volatility	↓	Ang, Hodrick, Xing, and Zhang (2006)
dinv	Inventory Changes	↓	Thomas and Zhang (2002)
ig	Investment Growth	↓	Xing (2008)
rev	Long-Term Reversal	↓	De Bondt and Thaler (1985)
me	Market Equity	↓	Banz (1981)
mdr	Maximum Daily Return	↓	Bali, Cakici, and Whitelaw (2011)
mom	Momentum	↑	Fama and French (1996)
ndf	Net Debt Financing	↓	Baker and Wurgler (2006)
ndm	Net Debt to Market	↓	Penman *et al.* (2007)
nef	Net Equity Financing	↓	Baker and Wurgler (2006)
nxf	Net External Financing	↓	Baker and Wurgler (2006)
noa	Net Operating Assets	↓	Hirshleifer *et al.* (2004)
npy	Net Payout Yield	↑	Boudoukh, Michaely, Richardson, and Roberts (2007)
o	O-Score	↓	Ohlson (1980); Dichev (1998)
oa	Operating Accruals	↓	Sloan (1996)
ocm	Operating Cash Flow to Market	↑	Desai, Rajgopal, and Venkatachalam (2004)
ope	Operating Efficiency	↑	Fama and French (2015)
ol	Operating Leverage	↑	Novy-Marx (2011)
rer	Real-Estate Ratio	↑	Titman *et al.* (2004)
rs	Relative Strength	↑	Jegadeesh and Livnat (2006)
rmom	Residual Momentum	↑	Blitz, Huij, and Martens (2011)
roa	Return on Assets	↑	Balakrishnan, Bartov, and Faurel (2010)
roe	Return on Equity	↑	Hou, Xue, and Zhang (2014)
sm	Sales to Market	↑	Barbee Jr, Mukherji, and Raines (1996)

tur	Share Turnover	↓	Datar, Naik, and Radcliffe (1998)
sue	Standardized Unexpected Earnings	↑	Foster, Olsen, and Shevlin (1984)
tbi	Taxable Income to Book Income	↓	Lev and Nissim (2004)

Table 7: Sorting variables in the factor library. The direction column indicates whether the long-short factor return goes long top-minus-bottom (↑) or bottom-minus-top (↓) of the characteristic distribution.

5.2. Access

Researchers retrieve precomputed series without rebuilding the pipeline. **tidyfinance** hosts the library on Hugging Face and exposes it through `download_data()`. Hugging Face retains a full commit history of each repository in the same way GitHub does, so legacy versions of the factor library remain accessible by referencing earlier snapshots; users can therefore reproduce results computed against an older release of the library even after the current version has been updated. An interactive dashboard at factors.tidy-finance.org provides the same series through a browser interface. The companion Python package **tidyfinance** offers an identical API for users who prefer Python.

A default call retrieves a single factor by name and lag:

```
download_data(
  domain = "tidyfinance",
  dataset = "factor_library",
  sorting_variable = "ag",
  sorting_variable_lag = "6m"
)
```

```
# A tibble: 779 x 17
       id date       ret_type   ret sorting_variable min_size_quantile
    <int> <date>     <chr>    <dbl> <chr>                          <dbl>
 1 297554 1960-02-01 vw           0 ag                               0.2
 2 297554 1960-03-01 vw           0 ag                               0.2
 3 297554 1960-04-01 vw           0 ag                               0.2
 4 297554 1960-05-01 vw           0 ag                               0.2
 5 297554 1960-06-01 vw           0 ag                               0.2
 6 297554 1960-07-01 vw           0 ag                               0.2
 7 297554 1960-08-01 vw           0 ag                               0.2
 8 297554 1960-09-01 vw           0 ag                               0.2
 9 297554 1960-10-01 vw           0 ag                               0.2
10 297554 1960-11-01 vw           0 ag                               0.2
# i 769 more rows
# i 11 more variables: exclude_financials <lgl>, exclude_utilities <lgl>,
#   exclude_negative_earnings <lgl>, sorting_variable_lag <chr>,
```

```
#   rebalancing <chr>, n_portfolios_main <chr>, sorting_method <chr>,
#   breakpoints_min_size_threshold <dbl>, n_portfolios_secondary <dbl>,
#   breakpoints_exchanges <chr>, weighting_scheme <chr>
```

The defaults passed to `dataset = "factor_library"` reflect one common construction choice but may not suit every research question. Verify that the selected combination matches the intended design before drawing conclusions.

Setting `fill_all = TRUE` overrides the default values for all unspecified preprocessing parameters and returns long-short returns for every combination of size cutoff, exclusion filter, rebalancing frequency, sorting method, number of portfolios, breakpoint exchange, and weighting scheme that matches the supplied filters. This call is the entry point for evaluating the robustness of a result across the full grid of plausible preprocessing steps. The call below returns a long data frame of monthly long-short returns for asset growth sort, with one row per specification-date pair and a specification `id` that maps back to the underlying construction choices.

```
ag_data <- download_data(
  domain = "tidyfinance",
  dataset = "factor_library",
  sorting_variable = "ag",
  fill_all = TRUE
)
```

The call returns 13,384,080 rows, covering 17,280 combinations of the choices in Table 6 that match the supplied filters. The `id` column encodes the construction choices for each specification; the `date` and `ret` columns give the monthly factor returns for that specification.

For finer control, **tidyfinance** exposes lower-level helpers that sit beneath the `download_data()` interface. `download_data("tidyfinance", "factor_library_grid")` returns the entire available data frame of factor construction specifications from our Hugging Face repository; `download_data("tidyfinance", "factor_library", ids = selected_ids)` retrieves returns for a specific vector `selected_ids` of precomputed specification IDs. Researchers who want to iterate over a custom subset of specifications rather than filter by parameter value will prefer this route.

```
full_grid <- download_data("tidyfinance", "factor_library_grid")

selected_ids <- full_grid |>
  filter(
    sorting_method == "univariate",
    sorting_variable_lag == "6m",
    rebalancing == "monthly",
    n_portfolios_main == 10,
    weighting_scheme == "VW",
    breakpoints_exchanges == "NYSE",
    min_size_quantile == 0.2,
    is.na(breakpoints_min_size_threshold),
    !exclude_financials,
    !exclude_utilities,
```

```
    !exclude_negative_earnings
  ) |>
  pull(id)

return_data <- download_data(
  "tidyfinance",
  "factor_library",
  ids = selected_ids
)
```

5.3. Construction-choice sensitivity

We illustrate the kind of sensitivity analysis the library supports by computing a Fama-French three-factor alpha for each specification of the asset-growth factor in `ag_data`. Because the factor return is the long-short payoff of a self-financing portfolio, it is already in excess form and the risk-free rate does not enter the regression. The Fama-French three factors are already available from the earlier `download_data(domain = "factors_ff", dataset = "Fama/French 3 Factors")` call in Section 3.3: `ff_factors` contains the monthly market excess return (`mkt_excess`), the size factor (`smb`), and the value factor (`hml`). For each specification i we estimate

$$r_{i,t} = \alpha_i + \beta_i \, \mathrm{MKT}_t^e + s_i \, \mathrm{SMB}_t + h_i \, \mathrm{HML}_t + \varepsilon_{i,t}$$

by ordinary least squares, where $r_{i,t}$ is the long-short return of specification i in month t. We report the intercept α_i as the abnormal return that the three-factor model leaves unexplained. Figure 5 plots the histogram of these alphas across all 17,280 specifications.

```
ag_alpha <- ag_data |>
  left_join(ff_factors, join_by(date)) |>
  group_by(id) |>
  reframe(estimate_model(
    pick(everything()),
    model = "ret ~ mkt_excess + hml + smb"
  )) |>
  select(id, alpha = intercept)

n_specs <- nrow(ag_alpha)
ag_alpha |>
  ggplot(aes(x = alpha)) +
  geom_histogram(binwidth = 0.001) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  scale_y_continuous(labels = label_comma()) +
  labs(
    x = "Alpha",
    y = "Count",
    title = paste(
      "Asset-growth premium across",
      format(n_specs, big.mark = ","),
```

```
    "specifications"
  ),
  subtitle = "FF3 alpha"
)
```

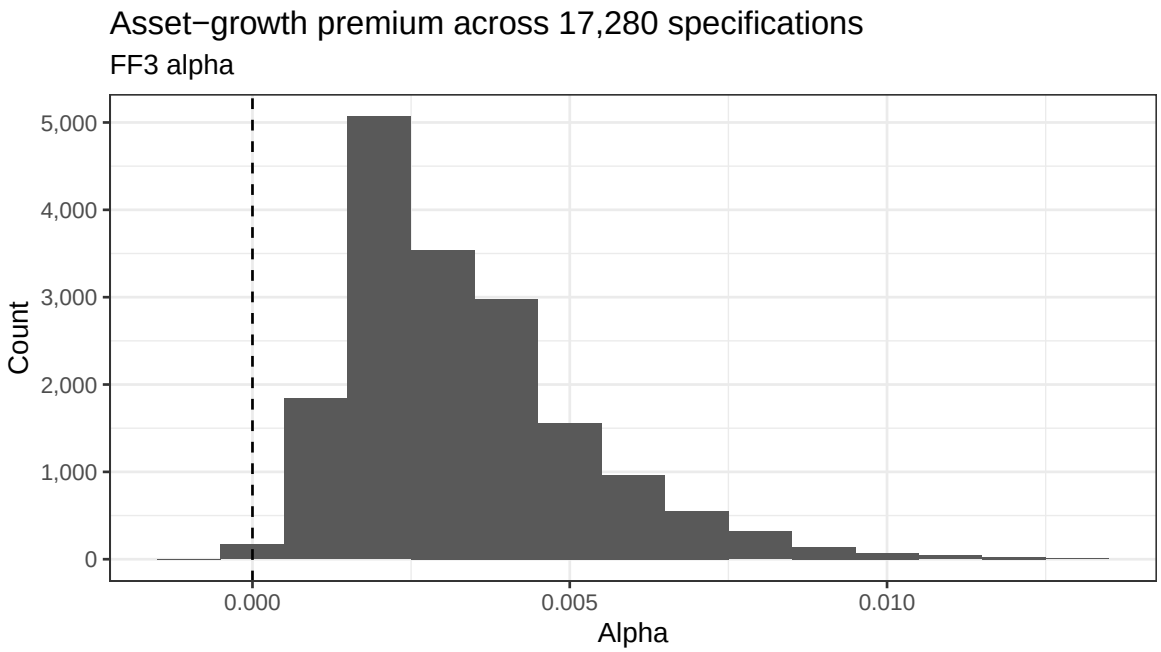


Figure 5: Distribution of asset-growth Fama-French three-factor alphas across construction choices. The histogram shows monthly alphas from regressing each of the 17,280 long-short specifications of the asset-growth factor on the market, size, and value factors, estimated by OLS. The dashed vertical line marks zero. Specifications vary as listed in Table 6.

6. Summary

Empirical asset pricing depends on dozens of construction choices that no consensus pins down. **tidyfinance** makes those choices first-class objects in the workflow: each is an argument, each has a documented default, each can be varied without rebuilding the pipeline. Disagreements over construction become empirical questions, not implicit assumptions buried in code.

We will maintain and extend the package on CRAN. The current release covers US equities through CRSP and Compustat; the next milestones add international equity panels, additional factor methodologies, and expanded fixed-income coverage beyond TRACE.

Beyond this paper, **tidyfinance** is part of a broader effort toward reproducible quantitative finance and more effective teaching of empirical methods. Companion materials at tidy-finance.org integrate the package into open coursework and self-study tracks for graduate finance and asset pricing.

Computational details

We obtained the results in this paper using R 4.5.2 with the following packages: **tidyfinance** 0.5.0.9002, **dplyr** 1.2.0, **tidyr** 1.3.1, **ggplot2** 4.0.0, **lubridate** 1.9.4, **arrow** 21.0.0.1, and **knitr** 1.51. CRAN hosts R and all packages at https://CRAN.R-project.org/. **renv** 1.1.5 ensures full reproducibility of the computational environment. Reproducing the factor return data requires an institutional subscription to WRDS, as the underlying CRSP and Compustat data are proprietary and cannot be redistributed.

Reproducing this paper takes about 15 minutes on a regular workstation. Recomputing the full factor library from raw WRDS data across all 841,536 specifications takes about 3 hours on a 64-core machine.

We publish all code, data workflows, and documentation at https://github.com/tidy-finance/jss-multilingual-factor-library. The code is released under the MIT License, which is compatible with the GNU General Public License (GPL).

```
old_width <- options(width = 65)
print(sessionInfo())
```

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: Europe/Copenhagen
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods
[7] base

other attached packages:
[1] lubridate_1.9.4        tidyr_1.3.1
[3] dplyr_1.2.0            tidyfinance_0.5.0.9002
[5] patchwork_1.3.2        scales_1.4.0
[7] ggforce_0.5.0          ggplot2_4.0.0

loaded via a namespace (and not attached):
 [1] utf8_1.2.6         sandwich_3.1-1     rappdirs_0.3.3
```

```
 [4] generics_0.1.4     renv_1.1.5          stringi_1.8.7
 [7] lattice_0.22-7     hms_1.1.4           digest_0.6.37
[10] magrittr_2.0.4     evaluate_1.0.5      grid_4.5.2
[13] timechange_0.3.0   RColorBrewer_1.1-3  fastmap_1.2.0
[16] blob_1.3.0         jsonlite_2.0.0      RPostgres_1.4.9
[19] DBI_1.2.3          purrr_1.1.0         frenchdata_0.2.0
[22] tweenr_2.0.3       codetools_0.2-20    httr2_1.2.2
[25] cli_3.6.5          rlang_1.1.7         dbplyr_2.5.2.9000
[28] polyclip_1.10-7    bit64_4.6.0-1       withr_3.0.2
[31] yaml_2.3.10        tools_4.5.2         tzdb_0.5.0
[34] curl_7.0.0         assertthat_0.2.1    vctrs_0.7.2
[37] R6_2.6.1           zoo_1.8-15          lifecycle_1.0.5
[40] stringr_1.5.2      bit_4.6.0           MASS_7.3-65
[43] arrow_21.0.0.1     pkgconfig_2.0.3     pillar_1.11.1
[46] gtable_0.3.6       glue_1.8.0          xfun_0.53
[49] tibble_3.3.0       tidyselect_1.2.1    knitr_1.51
[52] farver_2.1.2       htmltools_0.5.8.1   labeling_0.4.3
[55] rmarkdown_2.29     compiler_4.5.2      S7_0.2.0
```

```
options(old_width)
```

Acknowledgments

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Internet Appendix

IA-1. Variable construction

In this section, we show the construction details for the risk factors. We group them by their underlying mechanism as shown in Table 7 in the Appendix of the main paper.

IA-1.1. Financing

CDI - Composite Debt Issuance

Following Lyandres *et al.* (2008), we compute composite debt issuance as the logarithmic growth rate of book debt (i.e., current debt, DLC, plus long-term debt, DLTT) over the last five years, i.e.,

$$\text{CDI}_t = \log(\text{DLC}_t + \text{DLTT}_t) - \log(\text{DLC}_{t-60} + \text{DLTT}_{t-60})$$

We replace the missing values of DLC and DLTT with zero individually. Additionally, the sum of DLC and DLTT must be larger than zero, and is set to missing otherwise.

CSI - Composite Share Issuance

Following Daniel and Titman (2006), we measure composite share issuance from CRSP data as the difference between the change in market equity (ME) (i.e., the shares outstanding, shrout, times the absolute value of their price, altprc, in million USD) and the cumulative log return (where r is a simple return) in each month from t to $t-60$, i.e.,

$$\text{CSI}_t = \log(ME_t/ME_{t-60}) - \sum_{\tau=t-59}^{t} \log(1+r_\tau)$$

DBE - Change in Common Equity

Following Richardson *et al.* (2005), we calculate the change in common equity as the change in common equity (CEQ) over one year scaled by lagged assets (AT), i.e.,

$$DBE_t = \frac{CEQ_t - CEQ_{t-12}}{AT_{t-12}}$$

DCOL - Change in Current Operating Liabilities

Following Richardson *et al.* (2005), we measure the change in current operating liabilities as the change in current operating liabilities (i.e., current liabilities, LCT, minus current debt, DLC) over one year scaled by lagged assets (AT), i.e.,

$$DCOL_t = \frac{(LCT_t - DLC_t) - (LCT_{t-12} - DLC_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC with 0.

DFNL - Change in Financial Liabilities

Following Richardson *et al.* (2005), we measure the change in financial liabilities (i.e., the sum of current debt, DLC, plus long-term debt, DLTT, plus preferred stocks, PSTK) over one year scaled by lagged assets (AT), i.e.,

$$DFNL_t = \frac{(DLC_t + DLTT_t + PSTK_t) - (DLC_{t-12} + DLTT_{t-12} + PSTK_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC, DLTT and PSTK with 0, if at least one of them is given.

NDF - Net Debt Financing

Following Bradshaw, Richardson, and Sloan (2006), we measure net debt financing as cash proceeds from the issuance of long-term debt (DLTIS) minus cash payments for long-term debt reductions (DLTR) plus net changes in current debt (DLCCH) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NDF_t = \frac{DLTIS_t - DLTR_t + DLCCH_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

We replace the missing values of DLCCH with zero.

NEF - Net Equity Financing

Following Bradshaw *et al.* (2006), we measure net equity financing as proceeds from the sale of common and preferred stocks (SSTK) minus payments for the repurchase of common and preferred stocks (PRSTKC) minus cash payments for dividends (DV) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NEF_t = \frac{SSTK_t - PRSTKC_t - DV_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

NXF - Net External Financing

Following Bradshaw *et al.* (2006), we measure net external financing as the sum of net debt financing (NDF) and net equity financing (NEF) as defined above.

IA-1.2. Intangibles

HR - Hiring Rate

Following Belo *et al.* (2014), we measure the hiring rate as the change in the number of employees (EMP) over one year scaled by the average number of employees, i.e.,

$$HR_t = \frac{EMP_t - EMP_{t-12}}{\frac{1}{2}(EMP_t + EMP_{t-12})}$$

We exclude firms with a hiring rate of zero.

OL - Operating Leverage

Following Novy-Marx (2011), we measure operating leverage as cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA), both scaled by current total assets (AT), i.e.,

$$OL_t = \frac{COGS_t + XSGA_t}{AT_t}$$

RER - Real-Estate Ratio

Following Tuzel (2010), we measure the real-estate ratio in two subperiods differently.

(a) Prior to 1983, the real-estate ratio is the sum of buildings (PPENB) and capital leases (PPENLS) scaled by net property, plant, and equipment (PPENT), i.e.,

$$RER_t = \frac{PPENB_t + PPENLS_t}{PPENT_t}$$

(b) After the end of 1983, the real-estate ratio is the sum of buildings at cost (FATB) and leases at cost (FATL), both divided by gross property, plant, and equipment (PPEGT), i.e.,

$$RER_t = \frac{FATB_t + FATL_t}{PPEGT_t}$$

Subsequently, we winsorise the real-estate ratios in each fiscal year at the first and 99th percentile. The industry-adjusted real-estate ratio is obtained by subtracting the industry average real-estate ratio from each stock-specific real-estate ratio. We use 2-digit SIC codes to assign stocks to industries. We always require at least five observations to calculate the industry average each year.

IA-1.3. Investment

ACI - Abnormal Corporate Investment

Following Titman *et al.* (2004), we measure abnormal corporate investments using the variable CE, which is capital expenditures (CAPX) scaled by sales (SALE), then ACI is given by CE scaled by CE from the previous three years minus one, i.e.,

$$ACI_t = \frac{CE_t}{\frac{1}{3}(CE_{t-12} + CE_{t-24} + CE_{t-36})} - 1$$

We follow Hou *et al.* (2020) and exclude stocks with sales below 10 million dollars.

AG - Asset Growth

Following Cooper *et al.* (2008), we measure asset growth as the change in total assets (AT) over one year scaled by lagged assets (AT), i.e.,

$$AG_t = \frac{AT_t - AT_{t-12}}{AT_{t-12}}$$

DINV - Inventory Changes

Following Thomas and Zhang (2002), we measure the change in inventory as the annual change in inventories (INVT) scaled by average total assets (AT) over the current and the last year, i.e.,

$$DINV_t = \frac{INVT_t - INVT_{t-12}}{\frac{1}{2}(AT_t + AT_{t-12})}$$

DNOA - Change in Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, the change in net operating assets is the difference between net operating assets of fiscal year t and fiscal year $t-12$ scaled by lagged assets (AT), i.e.,

$$DNOA_t = \frac{NOA_t^{int} - NOA_{t-12}^{int}}{AT_{t-12}}$$

DPIA - Change in Property, Plant, Equipment, and Inventory

Following Lyandres *et al.* (2008), we measure the change in property, plant, equipment, and inventory by combining the change in gross property, plant, and equipment (PPEGT) with the annual change in inventory (INVT) and scaling this sum by lagged total assets (AT), i.e.,

$$DPIA_t = \frac{(PPEGT_t - PPEGT_{t-12}) + (INVT_t - INVT_{t-12})}{AT_{t-12}}$$

DWC - Change in Net Non-Cash Working Capital

Following Richardson *et al.* (2005), we measure non-cash working capital as

$$WC_t = (ACT_t - CHE_t) - (LCT_t - DLC_t)$$

where ACT corresponds to current assets, CHE to cash, LCT to current liabilities, and DLC to short-term debt. We set missing values of DLC to zero. Then, the change in net non-cash

working capital corresponds to the change in WC from fiscal year $t-12$ to fiscal year t scaled by lagged total assets (AT), i.e.,

$$DWC_t = \frac{WC_t - WC_{t-12}}{AT_{t-12}}$$

IG - Investment Growth

Following Xing (2008), we measure investment growth as the annual change in capital expenditures (CAPX) scaled by lagged capital expenditures, i.e.,

$$IG_t = \frac{CAPX_t - CAPX_{t-12}}{CAPX_{t-12}}$$

NOA - Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, net operating assets is the net operating assets of fiscal year t scaled by lagged assets (AT), i.e.,

$$NOA_t = \frac{NOA_t^{int}}{AT_{t-12}}$$

OA - Operating Accruals

Following Sloan (1996) (before 1988) and Hribar and Collins (2002) (after 1988), we measure operating accruals until 1988 as

$$OA_t = \frac{(\Delta ACT_t - \Delta CHE_t) - (\Delta LCT_t - \Delta DLC_t - \Delta TXP_t) - DP_t}{AT_{t-12}}$$

where ACT is current assets, CHE cash, LCT current liabilities, DLC short-term debt, TXP taxes payable, and DP depreciation and amortization. Moreover, we replace the missing values of DLC and TXP with zero. Δ represents changes over one year. From 1988 onwards, we measure operating accruals as

$$OA_t = \frac{NI_t - OANCF_t}{AT_{t-12}}$$

where NI is net income and OANCF corresponds to net cash flow from operations.

IA-1.4. Momentum

MOM - Return Momentum

Following Fama and French (1996), we measure return momentum from monthly CRSP returns (where r is a simple return) as the cumulative return from month $t-11$ to month $t-1$ skipping the most recent month, i.e.,

$$\text{MOM}_t = \exp\left(\sum_{\tau=t-11}^{t-1} \log(1+r_\tau)\right) - 1$$

RMOM - Residual Momentum

Following Blitz *et al.* (2011), we define the 11-month residual momentum (RMOM) in each month and for each stock as cumulative residual returns from month $t-11$ to month $t-1$, scaled by the standard deviation of residual returns over the same time horizon. Residual returns ϵ are obtained in each month from regressing monthly simple, excess stock returns from month $t-1$ to month $t-36$ on the Fama and French (1993) three-factor model. Throughout these rolling regressions, we always require 36 monthly returns.

RS - Revenue Surprise

Following Jegadeesh and Livnat (2006), we measure the change in revenues per share (DRPS) from quarterly Compustat data as quarterly sales (SALEQ) scaled by the product of quarterly shares outstanding (CSHPRQ) times their cumulative adjustment factor (AJEXQ), i.e.,

$$DRPS_q = \frac{SALEQ_q}{CSHPRQ_q \cdot AJEXQ_q} - \frac{SALEQ_{q-4}}{CSHPRQ_{q-4} \cdot AJEXQ_{q-4}}$$

Then, the revenue surprise corresponds to the change in revenues per share over the last four quarters scaled by the standard deviation of the change in revenues per share over the last eight quarters. We require at least six quarterly observations for this rolling standard deviation and remove infinite values. Finally, the earnings announcement date has to be after the fiscal quarter end date.[2]

SUE - Standardized Unexpected Earnings

Following Foster *et al.* (1984), we measure unexpected earnings per share (UES) from quarterly Compustat data as the change in split-adjusted earnings (which are equal to quarterly earnings per share, EPSPXQ, scaled by the correction factor for shares outstanding, AJEXQ) per share from its value four quarters ago

$$UES_q = \frac{EPSPXQ_q}{AJEXQ_q} - \frac{EPSPXQ_{q-4}}{AJEXQ_{q-4}}$$

Then, standardised unexpected earnings are defined as unexpected earnings per share divided by the standard deviation of unexpected earnings per share over the previous eight quarters.

[2]We follow the convention of Hou *et al.* (2020) and do not estimate a random walk model with drift to compute the surprise.

We require at least six quarterly observations for this rolling standard deviation. Finally, the earnings announcement date has to be after the fiscal quarter end date.[3]

52W - 52-Week High

Following George and Hwang (2004), we measure the 52-week high as month t's closing split-adjusted stock price scaled by the highest daily split-adjusted stock price over the previous twelve months. We only consider stock-month pairs with more than three valid price observations.

REV - Long-Term Reversal

Following De Bondt and Thaler (1985), we measure the long-term reversal effect by the cumulative returns from month $t-60$ to month $t-13$, i.e.,

$$\text{REV}_t = \exp\left(\sum_{\tau=t-60}^{t-13} \log(1+r_\tau)\right) - 1$$

IA-1.5. Profitability

CBOP - Cash-Based Operating Profitability

Following Ball *et al.* (2016), we measure cash-based operating profitability

$$\begin{aligned} CBOP_t = &(REVT_t - COGS_t - XSGA_t + XRD_t \\ &- \Delta RECT_t - \Delta INVT_t - \Delta XPP_t + \Delta DRC_t \\ &+ \Delta DRLT_t + \Delta AP_t + \Delta XACC_t)/AT_t \end{aligned}$$

where REVT is total revenue, COGS are cost of goods sold, XSGA are selling, general, and administrative expenses, and XRD are R&D expenses. Moreover, Δ refers to one-year changes, which we measure for accounts receivable (RECT), inventory (INVT), prepaid expenses (XPP), deferred revenues (the sum of current, DRC, and long-term, DRLT), trade accounts payable (AP), and accrued expenses (XACC). We set the missing values of XRD and all missing changes to zero.

GPA - Gross Profits to Assets

Following Novy-Marx (2013), we measure gross profits to assets as total revenues (REVT) minus cost of goods sold (COGS) scaled by current total assets (AT), i.e.,

$$GPA_t = \frac{REVT_t - COGS_t}{AT_t}$$

O - Ohlson's O-score

[3]We follow the convention of Hou *et al.* (2020) and do not estimate a random walk model with drift to compute the surprise.

Following Ohlson (1980), we measure O-score with the following linear relation

$$\begin{aligned} O_t = &- 1.32 - 0.407 \cdot \log(AT_t) + 6.03 \cdot \frac{DLC_t + DLTT_t}{AT_t} \\ &- 1.43 \cdot \frac{ACT_t - LCT_t}{AT_t} + 0.076 \cdot \frac{LCT_t}{AT_t} \\ &- 1.72 \cdot \mathbf{1}_{LT_t > AT_t} - 2.37 \cdot \frac{NI_t}{AT_t} - 1.83 \cdot \frac{PI_t + DP_t}{LT_t} \\ &+ 0.285 \cdot \mathbf{1}_{NI_t<0 \text{ and } NI_{t-12}<0} - 0.521 \cdot \frac{NI_t - NI_{t-12}}{|NI_t| + |NI_{t-12}|} \end{aligned}$$

where AT corresponds to total assets, DLC to short-term debt, DLTT to long-term debt, ACT to current assets, LCT to current liabilities, LT to total liabilities, PI to pretax income, DP to depreciation and amortization, and NI to net income. We winsorise all variables except for dummy variables at the first and 99th percentile of their respective distribution.

OPE - Operating Profits to Book Equity

Following Fama and French (2015), we measure operating profits to book equity as

$$OPE_t = \frac{REVT_t - COGS_t - XSGA_t - XINT_t}{BE_t}$$

where REVT corresponds to total revenues, COGS to cost of goods sold, XSGA to selling, general and administrative expenses, XINT to interest expenses, and BE to book equity. The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero. Moreover, missing values in COGS, XSGA, and XINT are set to zero.

ROA - Return on Assets

Following Balakrishnan *et al.* (2010), we measure return on assets as quarterly income before extraordinary items (IBQ) relative to lagged quarterly total assets (ATQ), i.e.,

$$ROA_q = \frac{IBQ_q}{ATQ_{q-1}}$$

Finally, the earnings announcement date has to be after the fiscal quarter end date.

ROE - Return on Equity

Following Hou *et al.* (2014), we measure return on equity as quarterly income before extraordinary items (IBQ) relative to lagged book equity (BEQ)

$$ROE_q = \frac{IBQ_q}{BEQ_{q-1}}$$

Quarterly book equity (BEQ) is computed as the (a) book equity of shareholders plus (b) balance sheet deferred taxes and investment tax credit minus (c) the book value of preferred stock. Regarding (a), we measure book equity of shareholders by shareholders' equity (SEQQ), or the sum of common equity (CEQQ) and the par value of preferred stock (PSTKQ), or if all previous items are unavailable by total assets (ATQ) minus total liabilities (LTQ). Regarding (b), the balance sheet deferred taxes and investment tax credit is TXDITCQ, or TXDBQ, or zero if both are missing. Regarding (c), the book value of preferred stocks corresponds to PSTKRQ, or PSTKQ, or zero if both are unavailable. Finally, the earnings announcement date has to be after the fiscal quarter end date.

TBI - Taxable Income to Book Income

Following Lev and Nissim (2004), we measure taxable income to book income as pretax income (PI) scaled by net income (NI)

$$TBI_t = \frac{PI_t}{NI_t}$$

We require positive pretax and net income.

IA-1.6. Size

ME - Size

Following Fama and French (1992), we measure size as the natural logarithm of the market equity. We obtain market equity data from CRSP by multiplying the shares outstanding (shrout * 1,000, as they are given in thousands) with the corresponding share price (the absolute value of altprc), i.e.,

$$ME_t = \log\left(shrout_t * 1000 * |altprc_t|\right)$$

IA-1.7. Trading frictions

AMI - Amihud Illiquidity Measure

Following Amihud (2002), we measure the illiquidity from daily return-to-volume (RV), which is the absolute daily return (r) scaled by the daily dollar trading volume (product of vol and prc), i.e.,

$$RV_t = \frac{|r_d|}{prc_t * vol_t}$$

Then, the Amihud illiquidity measure corresponds to the average return to volume estimate over the last six months. We require at least 50 observations for this average and adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

BETA - Beta Relative to the Market

Following Fama and MacBeth (1973), we measure the market beta β_1 from monthly CRSP data. Specifically, we run the following time series regression over the previous five years

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + u_t$$

Moreover, we require at least 24 monthly observations for the regression above.

DTV - Dollar Trading Volume

Following Brennan *et al.* (1998), we measure dollar trading volume from daily CRSP data as the average dollar trading volume (product of vol and prc) from month $t-6$ to month $t-1$ (with at least 50 observations). Moreover, we adjust dollar trading volume from NASDAQ according to Gao and Ritter (2010).

ISKEW - Idiosyncratic Skewness Relative to the Fama and French (1993) three-factor model

Following Bali *et al.* (2016), we regress daily excess returns in each month on the three Fama and French (1993) factor model (with a minimum of 15 daily observations), i.e.,

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

Then, idiosyncratic skewness is measured as the skewness of residuals u_t.

IVOL - Idiosyncratic Volatility Relative to the Fama and French (1993) model

Following Ang *et al.* (2006), we compute idiosyncratic volatility relative to the Fama and French (1993) three-factor model as the volatility of residuals from the following regression (with a minimum of 15 daily observations)

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

MDR - Maximum Daily Return

Following Bali *et al.* (2011), we measure the maximum daily (excess) return in each month from daily CRSP data (with at least 15 return observations).

TUR - Share Turnover

Following Datar *et al.* (1998), we measure the daily share turnover of each stock as the number of shares traded (vol) scaled by the number of shares outstanding (shrout), i.e.,

$$T_t = \frac{vol}{shrout}$$

Then, the share turnover for each firm in each month t is the average daily share turnover over the previous six months (with at least 50 observations). Moreover, we adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

IA-1.8. Valuation

BM - Book Equity to Market Equity

Following Davis, Fama, and French (2000), we measure the book-to-market ratio as book equity (BE) from Compustat divided by market equity (ME) from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$BM_t = \frac{BE_t}{ME_t}$$

The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero.

CFM - Cash Flow to Market Equity

Following Lakonishok *et al.* (1994), we measure the cash-flow-to-price ratio as income before extraordinary items (IB) plus depreciation (DP) scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$CFM_t = \frac{IB_t + DP_t}{ME_t}$$

We exclude all stocks with negative cash flows.

DM - Debt to Market Equity

Following Bhandari (1988), we measure the debt-to-market ratio as short-term debt (DLC) plus long-term debt (DLTT) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$DM_t = \frac{DLC_t + DLTT_t}{ME_t}$$

We exclude stocks with missing DLC and DLTT observations.

EBM - Enterprise Book Equity to Market Equity

Following Penman *et al.* (2007), we measure enterprise book equity scaled by market equity as net debt plus book equity scaled by net debt plus market equity

$$EBM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + BE_t}{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments, and ME to the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD). Lastly, in this case, book equity BE is computed as common equity (CEQ) plus TSTKP minus DVPA. Missing observations in DVPA and TSTKP are set to zero. Additionally, we require that the sum of net debt and book equity, as well as the sum of net debt plus market equity, are positive.

EM - Earnings to Market Equity

Following Basu (1983), we measure the earnings-to-price ratio as income before extraordinary items (IB) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$EM_t = \frac{IB_t}{ME_t}$$

We exclude firms with negative earnings.

NDM - Net Debt to Market Equity

Following Penman *et al.* (2007), we measure net debt to price as net debt from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$NDM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t}{ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments. Lastly, missing observations in DVPA and TSTKP are set to zero.

NPY - Net Payout Yield

Following Boudoukh *et al.* (2007), we measure the net payout yield as net payouts from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$\begin{aligned} NPY_t = &[(DVC_t + PRSTKC_t + \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV<0}) \\ &- (SSTK_t - \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV>0})]/ME_t \end{aligned}$$

where DVC are dividends from common stock, PRSTKC is the purchase of common and preferred stock, PSTKRV is the value of the net number of preferred stocks outstanding, and SSTK reflects the sale of common and preferred stocks. $1_{\Delta PSTKRV<0}$ is a dummy variable that has value one if the annual change in PSTKRV is negative and zero otherwise. Moreover, we exclude stocks with negative net payouts.

OCM - Operating Cash Flow to Market Equity

Following Desai *et al.* (2004), we measure the ratio of operating cash flows to price as operating cash flows (OC) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$OCM_t = \frac{OC_t}{ME_t}$$

Before 1988, we measure OC as funds from operations (FOPT) minus the change in working capital (item WCAP). Thereafter, we measure OC as net cash flows from operating activities (OANCF). Moreover, we exclude firms with negative operating cash flows.

SM - Sales to Market Equity

Following Barbee Jr *et al.* (1996), we measure the sales-to-price ratio (SM) as sales (SALE) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$SM_t = \frac{SALE_t}{ME_t}$$

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